Analysis of the Effects of Housing Permits on Population Growth from 2010-2022: Forecasting Models

Introduction

Population growth is one of the most important factors reflecting how communities change over time, as it can affect (and be affected by) labor market trends, infrastructure quality, public services, and demand for housing, making it crucial to understand the factors that have the strongest effects on population growth. One of the most well studied variables is the availability of housing, as the availability of housing can not only affect the price of housing (thereby affecting the cost of living) but also the physical capacity for a city to hold more people and hire more people within the area itself. House permits, which are a very strong correlate of housing construction, give a clear indication of changes in housing supply, and can be used to examine the relationship between construction and population growth. Additionally, house permitting is affected by a multitude of factors, ranging from the amount of land left for zoning, to the regulatory environment in the area. However, population growth is affected by many socioeconomic, demographic, and labor market factors, making the relationship between permitting and population growth far too complicated for simple statistical modeling.

The goal of this project is to evaluate the relationship between house permitting (a strong correlate of house construction) and population growth across U.S. counties from 2010-2022. Using county-level housing, socioeconomic, demographic, and labor market data, this project applies multiple statistical approaches, and machine learning methods to determine the relationship from different perspectives, including linear relationships, nonlinear prediction methods, growth forecasting, and causal inference. This specific report uses forecasting to

determine whether house permitting and historic population growth data can be used to predict future changes in population growth. The modeling focuses on the temporal relationship between historic population growth, house permitting, and present year population changes, while judging the accuracy of the resulting population forecasts.

Data

In order to determine the effect of housing permits on population growth, there were many covariates that were used for different reasons. These included the total permit rate per 1,000 people (obtained from the total number of permits in the county, divided by the population, multiplied by 1,000), as well as the number of single family permits per 1,000, the number of multifamily permits per 1,000 (specifically 2-4 unit permits), and the number of permits for buildings with at least 5 units (per 1,000). Each permit-rate variable was used as the main independent variable for its own regression, while the total permit rate was used for every other type of model. Additionally, the remaining housing variables included county-level homeownership rates, the growth rate in occupied housing units (which was lagged by a year), rent burden (the percentage of people paying more than 30% of their income in rent), and median house value. The labor market and economic characteristics that were used included median household income and the employment population ratio. The demographic variables that were used were the percent of people in a county who are white, the percent of people in a county who are black, the percent of people born outside the US, the change in the percent of people born outside the US (essentially functioning as net migration), as well as the county's median age. Lastly, I also used population density for the counties, which functions as a ceiling on how many people can live in an area before geographic constraints inhibit housing construction (albeit, for

many counties, that limit hasn't been reached). After finding the correct specifications for all of the variables, I also added two interaction terms, permits multiplied by housing growth, and permits multiplied by median house value.

Data Preparation

With the exception of the permit rate, I gathered all of the data from the American Community Survey's 5 year estimates. I used 5 year estimates instead of 1 year estimates because ACS data becomes very noisy when applied to small counties due to their small sample sizes for data collection. For the permit rate, I gathered information on total permits from the Department of Housing and Urban Development, which has the data from 1980-2022, after which I divided the number of each permit type by the population, and I multiplied them by 1,000 to get the per 1,000 rate. I used data from 2010-2022 for almost all of the variables because much of the ACS data starts in 2010, while the housing permit data set runs up to 2022. I also used 2023 data for population, foreign-born population, and the number of occupied housing units. These were used to create growth rates which were mostly used for lags so that I could use the 2022 data to determine trends. The data for the variables (from ACS data) was collected using API keys, and they were merged using SQL joins (mostly inner joins), with some variable transformations and calculations being performed in SQL using window functions, with other transformations being performed either in Excel or in Python workbooks.

For all of the models, I had to make some decisions about what data to include and what to remove. Firstly, since the COVID-19 pandemic falls within the time frame of this project, there are some years where population growth is significantly lower as a result of a crisis that can't easily be seen in the data. The period when this issue was most apparent was 2020-2021, as

population growth was significantly lowered from 2020 to 2021 as hundreds of thousands died from the virus. While even more people died in 2021, this ended up having less of a deleterious effect on population growth because of the substantial increases in immigration during 2021. Since corresponding population growth was the variable of choice, population growth from 2020 to 2021 was associated with the values for each of the variables in 2020. As a result of the issues that 2020 would present for the data, I made the decision to remove the 2020 data from consideration, as well as its corresponding population growth. Although this made models like forecasting far more difficult, the alternative was worse as a whole year's worth of data would be of incredibly poor quality.

Additionally, since there were some very noticeable outliers which could heavily alter the relationship between an independent and dependent variable, I tended towards removing them. This was firstly done by removing counties with a population below 2,500 as population growth figures are incredibly noisy at such low levels, since even a couple families deciding to move out can have a substantial effect on population growth. Additional outliers which were removed included counties which allowed more than 20 permits per 1,000 (a sample of roughly 30 observations), counties which had population growth rates above 8%, and below -5%, (a sample of roughly 40 observations), and counties which saw their number of available housing units either increase by more than 15% in one year, or decrease by more than 12.5% in one year (a sample of roughly 160 observations). While these removed a meaningful amount of data (roughly 0.6% of the data in the sample), even small amounts of outliers were capable of changing the best relationship between an IV and a DV from a linear relationship to a logarithmic one, which is especially problematic for something like regression modeling, where the wrong relationship being specified can make models perform far worse.

Variables

Besides the house permit rate, which is the main independent variable for the project, the housing variables in question were used to communicate how affordable housing is in some way, or to communicate the availability of housing in full. Median house value, rent burden, and homeownership rate are used for different reasons to tell how affordable housing is in terms of the price of housing, the household level constraints from owning housing, or how many people own homes (because many people who don't own houses would own them if they could). The growth in housing units is a reflection of the housing stock at any given time, and even though it could be highly correlated with permitting frequency, it isn't, even when it's lagged.

The employment population ratio was used because it's a simple reflection of how many people work and is superior to the employment rate in some respects because it can better account for an area having lots of discouraged workers, retirees, or limited employment opportunities. Median household income was used to determine how wealthy the average person is in an area.

When it came to demographic variables, I used the three groups (white people, black people, and foreign born people) for different reasons. There is a well documented history of white people moving out of urban areas and into suburban areas when they become more racially diverse, especially with respect to black people moving into an area, and this pattern has been present as suburbs have become more diverse as well (even in middle class neighborhoods), so a higher share of black people in an area may mean that white people move to a nearby county[1][2][3]. I used the white percentage to convey the level of racial homogeneity which may be a correlate of increased housing values for the same reason that an increased presence of minorities may be a correlate of decreased house values. I used the foreign born share not only as

a variable related to diversity, but also as a way to show a crude form of net migration which is itself a massive component of population growth that may not be captured by the data in any other way, especially relevant because the reason for immigrants coming to a certain county aren't likely to be properly predicted from the data. While immigrant flows are somewhat correlated with local labor market conditions, the nature of a labor market can only partially be inferred from the data in question[4].

Lastly, I used population density to communicate the extent to which a county has already been urbanized, its ability to get more permits approved, and its capacity to become more densely populated. I chose this as a measure instead of an urban vs rural percentage because the definition of an urban area has become more restrictive with time, including through an increase in the minimum population requirement.

Methodology

In order to make the proper models, it's critical to ensure that each variable is correctly specified. This is done by checking for the shape of each variable when plotted against population growth, along with the R^2 and the RMSE (root mean squared error) having a minor role in determining whether a transformation is best. However, if the relationship between a given variable and population growth looks linear, I defaulted to using a linear transformation to avoid the risk of fixating too much on small deviations in R^2 and RMSE. Since population growth is more or less normally distributed, it does not require a logarithmic transformation. Therefore, the other variables are measured against a dependent variable which has a linear specification. Ultimately, for most variables, a linear specification was adequate. This includes permits per 1,000 (which applies to all 4 of the permit variables), homeownership rate, the

employment population ratio, the change in occupied housing units, the change in the percent of foreign born people in a county, the rent burden, the median house value, and the median age in years, as well as the two interaction terms which used linear specifications for both of their variables. Logarithmic specifications were used for the foreign born population share, as well as population density. A quadratic specification was used for the percent of people in a county who are white, and a cubic specification was used for the percent of people in a county who are black, while a reciprocal specification was used for the median household income. The forecasting for this project was divided into two sections, one forecast with XGBoost, and one with standard ARIMA time series forecasting

XGBoost Forecast

When it came to performing forecasting with XGBoost, the data was divided into training and testing sets, with everything before 2016 being part of the train data, and everything afterwards being part of the test data. This forecasting model was used to determine growth trajectories for each quartile. As a result, the data was split into observations where a county stayed in the same permitting quartile for two consecutive years. Even though a lot of observations dropped out, there were still over 20,000 combined observations across the 4 quartiles, ranging from 4,717 in the third quartile (50th-75th percentile), to 6,218 in the first quartile (everything up to the 25th percentile). Since 2020 was previously eliminated from the dataset due to being the year where population growth numbers were most affected by the COVID pandemic, it was excluded from the model, and since this forecast required another population growth lag, 2019 was also excluded). After the data was obtained, four XGBoost gradient boosting models were performed (one for each quartile), with the parameters in question

being max_depth, learning_rate, n_estimators, subsample, colsample_bytree, min_child_weight, gamma, reg_alpha, and reg_lambda. To avoid redundancy, each of these parameters are explained in detail in my gradient boosting report that I wrote for the project. After performing hyperparameter tuning, I made graphs comparing the actual vs forecasted values for each value in the test data, followed by graphs comparing the actual population growth to the forecasted population growth.

ARIMA Forecast

ARIMA refers to Autoregressive Integrated Moving Average used for time series modeling, making it possible to forecast future values[5]. ARIMA uses autoregressive and moving average approaches to use the prior behavior of time series data to predict future values. However, it's very important for a time series forecast to be stationary, which is when the data's statistical properties are constant with time, including the mean and variance. An example of non-stationary data is the performance of an ascendant stock (like that of Nvidia). In that case, the mean is not constant because the price of the stock tends to increase. If non-stationarity is present, then the data has to be differenced until it becomes stationary, which occurs when the p-value is below 0.05 based on an augmented Dickey-Fuller test[6]. Stationary time series' are somewhat easy to predict because you can safely assume that their statistical properties will likely hold in the future because they either already did in the past, or they did after you differenced the data such that they would hold.

After adjusting for stationarity, you can determine the extent to which time series data is correlated with prior values by calculating the auto-correlation of the data, which helps determine whether the data is random, or how one observation relates to observations from

different points in time. Autocorrelations are sometimes plotted to see how the points correlate including the lagged unit. Autocorrelation coefficients make up the autocorrelation function (ACF). For ACF plots, the correlation coefficient is shown along the y-axis, while the number of lags are shown on the x-axis. Another function, and the one more relevant for this project was the partial autocorrelation function (PACF), which shows the relationship between y_t and y_{t-k} (essentially y of this year, vs y of a year beforehand in the context of this data, since it's yearly data). y_t and y_{t-1} might be correlated, but y_t might also be correlated with y_{t-2}, or y_{t-3}, and this may be because those two years give new information, but also because they may correlate with y_{t-1}. To mitigate this issue, partial autocorrelations help remove some lagged observations by evaluating the relationship between y_t and y_{t-k} after removing the effects of each lag from 1-k. The first partial autocorrelation is identical to the first autocorrelation (since there's nothing between the two values to have to fix), but after that, the effect of the PACF function starts to take effect. In the case of monthly or quarterly forecasts, a slew of prior months/quarters often show considerable correlations, but for yearly forecasts, that's not quite as common. As time goes on, the autocorrelations tend to decrease, since data from 1 year and 2 months is not likely to be as correlated as the data from 2 months ago.

In ARIMA modeling, autoregression refers to the variable of interest (population growth in this instance) being modeled using a linear combination of prior values of the variable, essentially being a regression of the dependent variable against past values of itself. On the other hand, moving average models use previous forecast errors. The value of y in the moving average component is essentially a weighted moving average of previous errors. Once the data has been made stationary, and the information for the auto-correlations has been found, there are three parameters needed to form an ARIMA model, known as p, d, and q. P is the order for the

autoregressive component of ARIMA, d is the number of differences that were needed, and q is the order for the moving average component, with the information often written as ARIMA(p, d, q).

Programming languages like Python have an ARIMA function that can be used to determine the three parameters. While looking at a typical plot of the data against time won't give you much information about the values of p and q, the ACF and PACF plots can be used to help identify them. AR terms may be needed in the model when ACF plots show ACF functions gradually decaying, approaching zero, with the PACF plot quickly approaching zero, and there being a positive value at lag 1. Meanwhile, AR terms may be needed when there is a negative autocorrelation at lag 1, ACF drops quickly after a couple lags, and PACF slowly decreases. In practice, the first two quartiles had positive values at lag 1, and quartiles 3 and 4 had values that were barely within the 95% confidence interval, meaning they weren't statistically different from zero but they were close. On the other hand, none of the patterns in the ACF and PACF data illustrated a clear need for MA components. Ultimately, only the first quarter needed a MA component, and all four of the quarters needed AR components. The final ARIMA model types were ARIMA(2, 0, 1), ARIMA(5, 0, 0), ARIMA(2, 0, 0), and ARIMA(2, 0, 0) respectively for the four quartiles. These models have AR values of at least 2 because they're used to map out complex models with many nonlinear terms. The way to determine the strength of each combination of p, d, and q is through examining the model's goodness of fit and model simplicity. In the case of this project, AIC was used, which is a composite of both, where lower values for AIC are better.

Results

XGBoost Forecast

After finding the right hyperparameters for each quartile, the plots comparing the actual values of population growth to the predicted values from the XGBoost forecast model show a somewhat strong fit. Additionally, the forecasts comparing actual vs predicted year on year population growth were quite similar, with the model worse at predicting the pattern in the later years. The models capture both the overall magnitude of population growth, and the fluctuations in population growth quite well, with the largest discrepancies in yearly population growth being at roughly 0.1 to 0.15 percentage points. The results for the XGBoost forecasts are shown below:

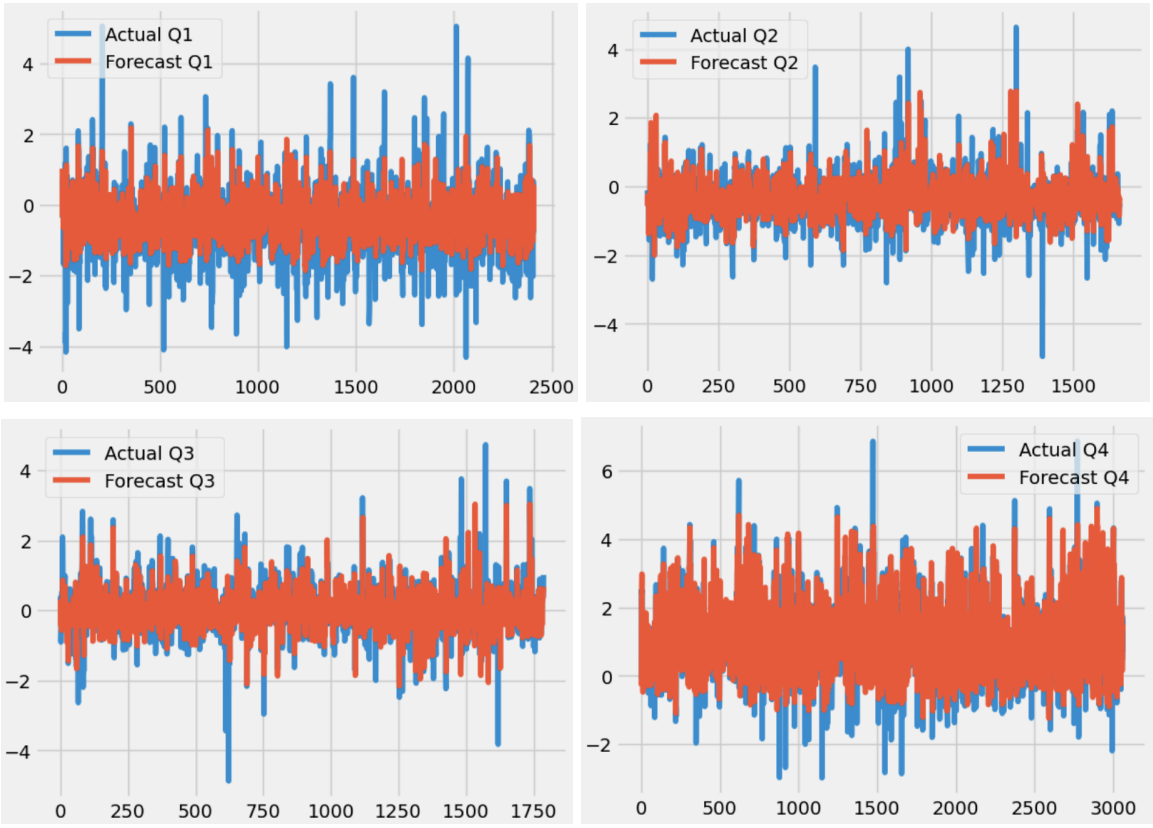


Figure 1: Plots comparing actual values of population growth to the predicted values of population growth for observations in each quartile.

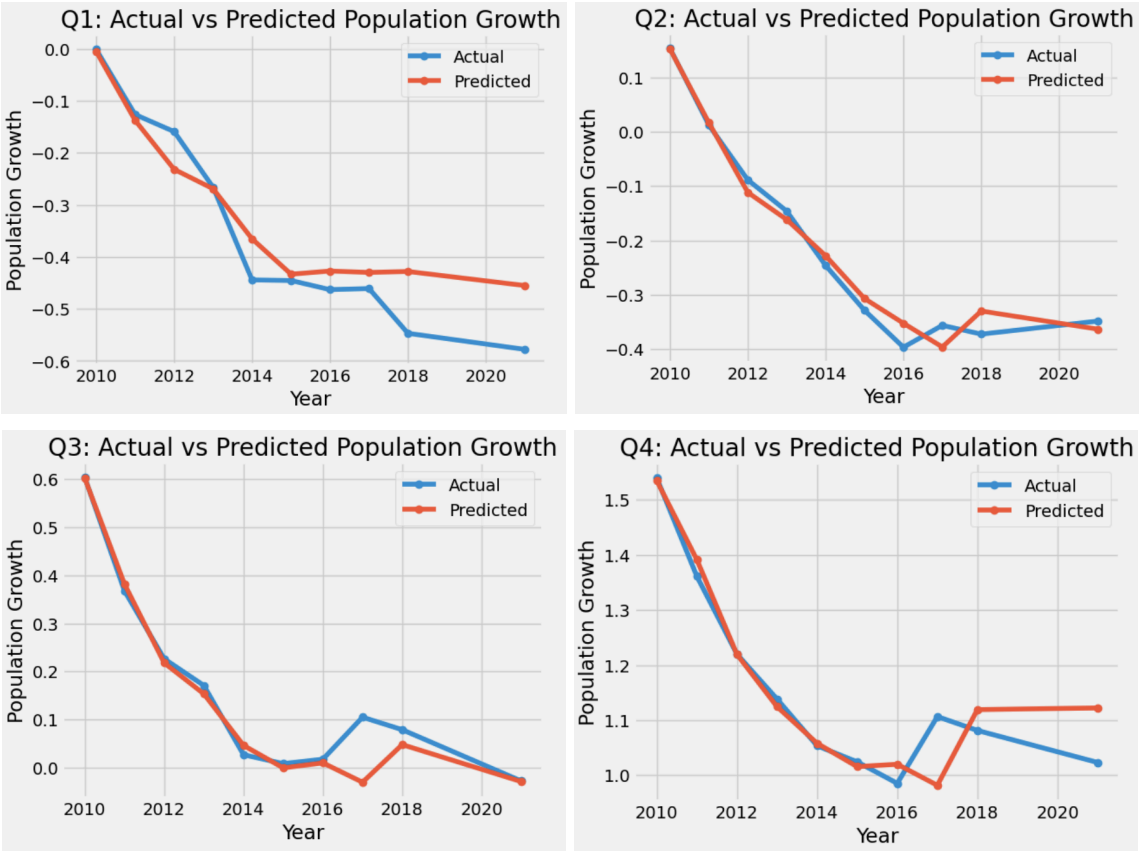


Figure 2: Plots comparing the actual and predicted population growth rates for each quartile from 2010-2021.

ARIMA Forecast

After having found the right combinations of p, d, and q for each quartile, and performing the forecasts, the forecast results generally exhibited a rebound in population growth. Although the population growth rates for each quartile decreased with time, the drop in population growth was most pronounced in the early years of the sample (with growth dropping by roughly 0.5 percentage points from 2010-2016), while the growth rates then mostly stabilized for the last few years. As a result, the population growth rates for each quartile increased in the next 5 years, especially for the second quartile. However, when performing a forecast on all counties put together, the opposite pattern emerged, as the overall population growth rate dropped from

roughly 0.4% to roughly 0% in the same 5 year period. Since the overall population growth forecast required an ARIMA(5, 2, 3) model and the data only covered 13 years, I evaluated the next best model, which was ARIMA(3, 2, 0), which performed very similarly. The forecasts for the overall population growth rate using ARIMA(5, 2, 3) and ARIMA(3, 2, 0) models, as well as the overlaid forecast containing all four permitting quartiles are shown below:

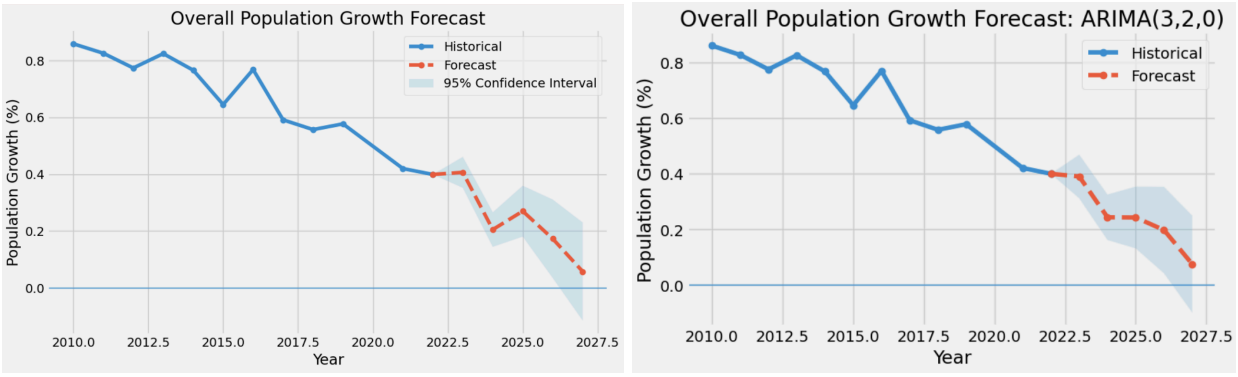


Figure 3: Overall Five Year Population Growth Forecasts for the Best Performing Model, and the Next Best Model Which Requires Fewer Autoregressive Components.

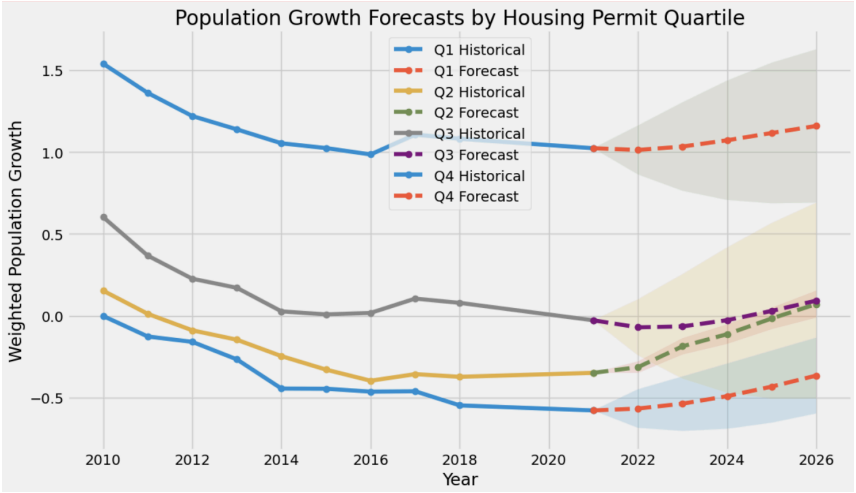


Figure 4: Five Year Population Growth Forecasts for Each Quartile of House Permitting.

Discussion

XGBoost Forecast Model

The results for the XGBoost forecasts indicate that increased house permitting is strongly associated with population growth, as each successive quartile with a higher permitting frequency corresponds to increased county-level population growth. The XGBoost models themselves were worse at predicting growth rates after 2016, but this was partially a result of the fact that the growth rates were less smooth and more unpredictable in later years, making the data intrinsically tougher to model.

The XGBoost models present some strengths because they model population growth using gradient boosting, with all of the nonlinear specifications of variables that were used for other models like causal inference and neural networks, while also having an additional variable for the growth rate from 2 years earlier, allowing it to capture complex relationships between variables quite well. In spite of the fact that population growth was less predictable in the later years of the sample, the model still performed exceptionally well, with no particularly large errors across any of the forecasts. However, in spite of the model performing very well, it cannot be used to infer causal relationships, but rather, just to predict future trends in population growth.

ARIMA Forecast Model

The final results show that an increase in house permitting is highly correlated with increased population growth. When comparing quartiles in the final year of the sample, the gap in population growth between the first and fourth quartile is 1.5 percentage points, and while every quartile's population growth forecast shows a rebounding effect in terms of population

growth, that gap remains intact. However, as a result of forecasting uncertainty, it is harder to determine whether the differences in population growth are truly meaningful. While the order of population growth rates remains the same with each successive quartile, the gap in growth rates shrinks with respect to quartiles 1-3, especially when it comes to quartiles 2 and 3, as the gap is neither statistically significant, nor easily noticeable to the naked eye. Even though the difference in growth rate between quartiles 1 and 3 remains similar, that gap is not statistically significant.

Conclusions

One of the most interesting takeaways from the results is that each quartile experiences a rebounding effect in terms of population growth, but the overall population growth forecast exhibits a significant decrease. Although the results seem contradictory, the reason why it happened is likely because those quartiles each showed a change in the rate at which the population declined, while the overall forecast didn't, as population growth steadily decreased. This difference may be a result of the fact that the forecast was based on counties which stayed in the same quartile for two consecutive years, which limits the sample size.

However, while the prior issue is easy to explain and understand, the main problem with the ARIMA forecasts (much more important than with XGBoost forecasts) is that the number of years analyzed is simply very limited, as there is only a 13 year time frame, and even that is reduced as a result of filtering out the 2020 corresponding population growth rates to avoid the issues from the main year affected by COVID. An ideal forecast would need at least 20 years of solid data, and probably more in order to help the ARIMA specifications have more information to learn from when predicting future values, therefore the takeaways from the ARIMA forecasting results are somewhat limited.

That being said, the overall takeaways remain very similar, because the XGBoost model had very similar patterns in terms of population growth rate for each quartile, showing that the correlation between increased house permitting and population growth remains quite strong, even though neither model can determine causality. The findings suggest that policies focused on increasing housing availability may help give counties the ability to reverse trends related to stagnant population growth, which has plagued many US counties for decades.

Appendix

The housing permits dataset was obtained from the department of housing and urban development[7]. The ACS data included the datasets for Selected Social Characteristics, Selected Housing Characteristics, Income in the Past 12 Months, Place of Birth by Poverty Status in the Past 12 Months in the United States, Age and Sex, Hispanic or Latino Origin by Race, Selected Economic Characteristics, and the Annual Geographic Information Table (GEOINFO)[8][9][10][11][12][13][14][15].

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